

MUHAMMAD MUNEER

Software Engineer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Software Engineer with full-stack experience (React, Node.js, Express, MongoDB) and hands-on AI/ML and Generative AI experience, including LLM applications, multi-agent systems, and Retrieval-Augmented Generation (RAG). Skilled in Python, FastAPI, PostgreSQL, LangChain, and LangGraph, with a track record of building production-oriented ML and backend systems during recent AI/ML and Data Engineering internships.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Machine Learning: Scikit-learn, CatBoost, Pandas, NumPy

Generative AI / LLM: LangChain, LangGraph, Google GenAI (Gemini), RAG, Vector Databases (pgvector), Hugging Face

Backend: FastAPI, Node.js, Express.js, REST APIs, JWT

Databases: PostgreSQL, MongoDB, SQL Server

Frontend: React.js, Next.js, HTML, CSS, Bootstrap, Tailwind CSS

Tools & Deployment: Git, GitHub, Postman, SSIS, Vercel, Render, Railway

EXPERIENCE

AI/ML Engineer Intern — IBEX

July 2026 – Present

IT Development

- Developed an LLM-powered chatbot using Python, FastAPI, LangChain, Google GenAI (Gemini), React, and PostgreSQL, supporting persistent chat and message management.
- Built a Salary Prediction ML pipeline with Python and Scikit-learn, covering preprocessing, model training, and evaluation, deployed via a FastAPI prediction endpoint.
- Developed an Activity Prediction model using CatBoost on historical employee activity data to forecast workforce activity patterns.
- Built a FastAPI + PostgreSQL agent adherence system that compares scheduled versus actual activity to classify agents as adherent or non-adherent.

Data Engineer Intern — IBEX

Feb 2026 – May 2026

IT Development

- Built ETL pipelines using SQL, SSIS, and CSV-based sources to extract, clean, and transform large-scale employee activity data for downstream analytics.
- Wrote and optimized SQL queries as part of data extraction and transformation workflows.

PROJECTS

Multi-Agent AI Assistant

Python, LangGraph, LangChain, Gemini

- Built a supervisor-based multi-agent system that routes user queries to specialized agents for weather, news, currency/commodities, country information, and research.
- Integrated Google Gemini for reasoning and tool-calling, connecting each agent to live external APIs for real-time data.
- Designed the agent orchestration and state-passing logic so the supervisor coordinates multiple agents through a single entry point.

RAG Assistant with Voice Interface

FastAPI, PostgreSQL (pgvector), Gemini

- Built a Retrieval-Augmented Generation (RAG) assistant using FastAPI, PostgreSQL with pgvector, and Google Gemini to answer questions over custom documents.
- Implemented a document ingestion pipeline covering chunking, embedding generation, and vector storage/retrieval.
- Added a voice interface for spoken input and output, extending the assistant beyond text-only interaction.

Cricket Tournament Management System

MERN, Socket.IO, JWT

- Developed a full-stack tournament management platform using React, Node.js, Express.js, and MongoDB with role-based access for admins, organizers, and team owners.
- Implemented JWT-based authentication with access/refresh tokens and secured the API with Helmet, CORS, and HPP.
- Added real-time updates using Socket.IO for live tournament and match information.

EDUCATION

Bachelor of Science in Software Engineering (BSSE)

Feb 2022 – Dec 2025

University of Karachi, Karachi, Pakistan | CGPA: 3.4 / 4.0

TRAINING / CERTIFICATIONS

- Web and App Development — course focused on contemporary web technologies and frameworks

LANGUAGES

Urdu — Native English — Intermediate